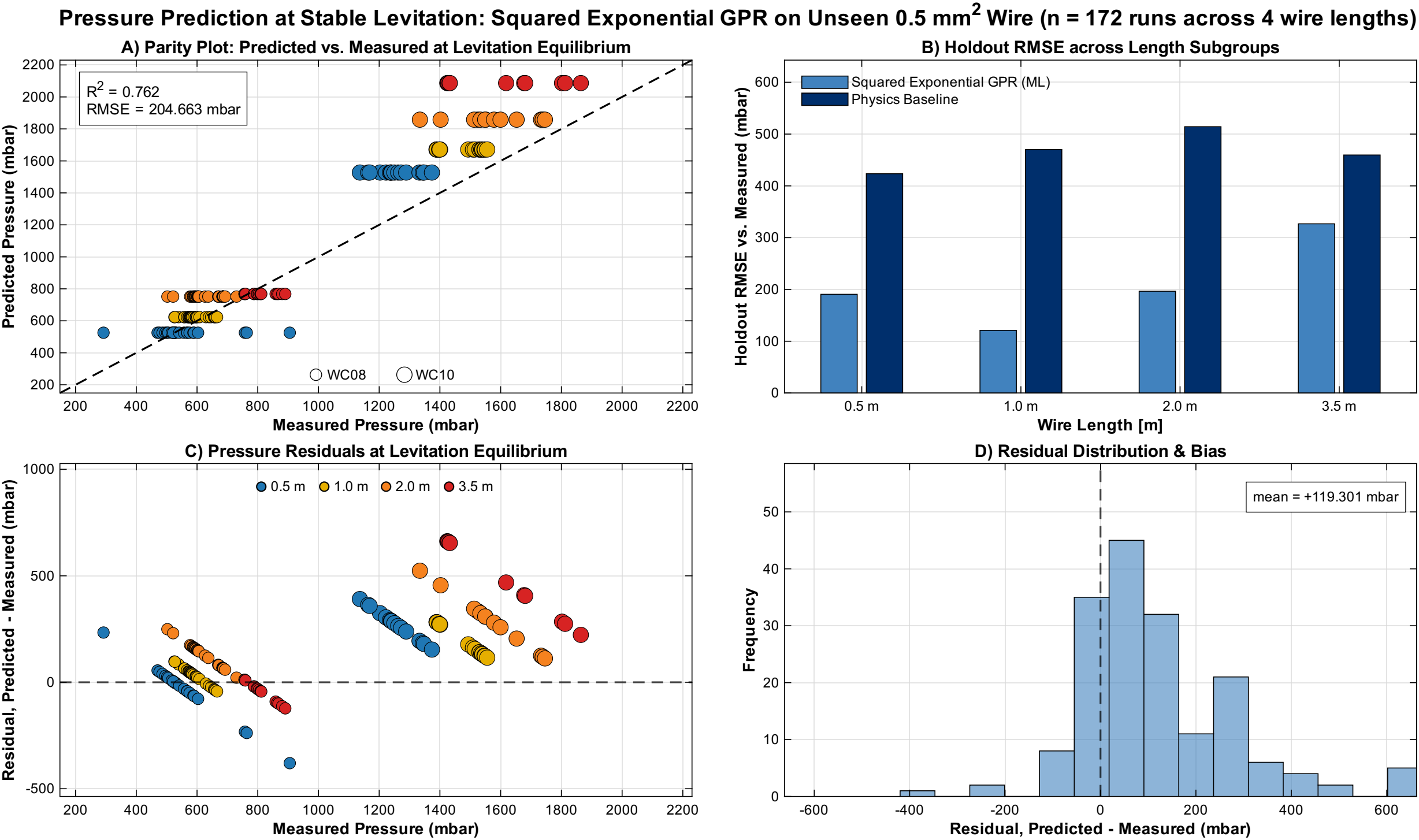

Pressure Prediction at Stable Levitation: Squared Exponential GPR on Unseen 0.5 mm² Wire (n = 172 runs across 4 wire lengths)
A) Parity Plot: Predicted vs. Measured at Levitation Equilibrium
R² = 0.762
RMSE = 204.663 mbar
WC08
WC10
Predicted Pressure (mbar)
Measured Pressure (mbar)
B) Holdout RMSE across Length Subgroups
Squared Exponential GPR (ML)
Physics Baseline
Holdout RMSE vs. Measured (mbar)
0.5 m
1.0 m
2.0 m
3.5 m
Wire Length [m]
C) Pressure Residuals at Levitation Equilibrium
0.5 m
1.0 m
2.0 m
3.5 m
Residual, Predicted - Measured (mbar)
Measured Pressure (mbar)
D) Residual Distribution & Bias
mean = +119.301 mbar
Frequency
Residual, Predicted - Measured (mbar)